Hyper Tech – Data Analytics & Business Intelligence Project

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Introduction

Hyper Tech is a leading electronics and computer hardware store specializing in high-performance devices, components, and accessories. As the company continues to grow, the need for accurate, data-driven decision-making becomes essential.
This project aims to build a complete Business Intelligence solution that transforms raw operational data into meaningful insights that support strategic planning and performance optimization.
Throughout the project, our team followed a full end-to-end data lifecycle — starting from data collection, cleaning, and quality assurance, all the way to building a structured data warehouse and extracting advanced insights using modern analytical tools.
The final deliverable is a fully interactive dashboard built with Power BI and Tableau that visualizes sales performance, customer behavior, product trends, and return impacts, enabling Hyper Tech's management to make smarter, faster, and more informed business decisions.
This presentation showcases the journey of developing this analytical system, the challenges we solved, and the value the final insights bring to the Hyper Tech store.

Data Collection

The Data Collection phase marked the beginning of building our analytical foundation for the Hyper Tech project. During this stage, we gathered all the essential raw data needed to analyze sales performance, customer behavior, product categories, and return patterns.

We collected multiple core datasets, including:
Customers Table – customer information and demographics
Territory Table – regional sales mappings
Product Table – complete product catalog
Categories Table – hierarchical product grouping
Calendar Table – generated time dimension
Returns Data – records of returned orders and financial impact

Building the Order Details Table
Originally, our data included a general Orders table. However, once we started modeling the relationships, we discovered a many-to-many issue between Orders and Products (one order can contain multiple products, and one product appears in multiple orders).

To solve this and ensure a proper star-schema model:
1- We created a new table called "OrderDetails"
2- We removed the original "Orders table"
3- We enhanced the new table and renamed it "FactOrderDetails"

Data Cleaning

The Data Cleaning Phase ensured that all collected datasets were accurate, consistent, and ready for analytical modeling. This step was essential for transforming the raw operational files into reliable data suitable for building the data warehouse and dashboards.

```sql
---------- Customers Table ----------

-- 1. Remove full empty customer row
DELETE FROM [Customers Table]
WHERE (EmailAddress IS NULL OR EmailAddress = '')
  AND (FirstName IS NULL OR FirstName = '')
  AND (LastName IS NULL OR LastName = '')
  AND (BirthDate IS NULL OR BirthDate = '');

----------------------------------------------------------------

-- 2. Fill missing emails with default value
UPDATE [Customers Table]
SET EmailAddress = 'Unknown'
WHERE EmailAddress IS NULL OR EmailAddress = '';

----------------------------------------------------------------

-- 3. Fix email typos (common “gamil” mistake)
UPDATE [Customers Table]
SET EmailAddress = REPLACE(EmailAddress, 'gamil.com', 'gmail.com')
WHERE EmailAddress LIKE '%gamil.com%';

----------------------------------------------------------------

-- 4. Trim extra spaces in names
UPDATE [Customers Table]
SET FirstName = LTRIM(RTRIM(FirstName)),
    LastName  = LTRIM(RTRIM(LastName));

----------------------------------------------------------------

-- 5. Fill missing names with default value
UPDATE [Customers Table]
SET FirstName = 'Unknown'
WHERE FirstName IS NULL OR LTRIM(RTRIM(FirstName)) = '';

UPDATE [Customers Table]
SET LastName = 'Unknown'
WHERE LastName IS NULL OR LTRIM(RTRIM(LastName)) = '';

----------------------------------------------------------------
```

```sql
---------- Calender Table ----------

-- 1. Remove duplicate dates (keep one)
WITH CTE AS (
    SELECT *,
        ROW_NUMBER() OVER(PARTITION BY Date
    FROM [Calendar Table]
)
DELETE FROM CTE WHERE rn > 1;
```

```sql
---------- Orders Table ----------

-- 1. Delete rows where all key columns are NULL
DELETE FROM [Orders Table]
WHERE (Order_ID IS NULL OR Order_ID = '')
  AND (Order_Date IS NULL OR Order_Date = '')
  AND (Customer_ID IS NULL OR Customer_ID = '')
  AND (Territory_ID IS NULL OR Territory_ID = '')

----------------------------------------------------------------

-- 2. Remove duplicated Orders
WITH CTE AS (
    SELECT *,
        ROW_NUMBER() OVER(PARTITION BY Order_ID ORDER BY Order_ID) AS rn
    FROM [Orders Table]
)
DELETE FROM CTE WHERE rn > 1;

----------------------------------------------------------------
```

```sql
---------- Products Table ----------

-- 1. Replace missing or NULL category ids with default
UPDATE [Product Table]
SET Category_ID = 0
WHERE Category_ID IS NULL;

----------------------------------------------------------------

-- 2. Fix invalid numeric values (non-numeric prices)
UPDATE [Product Table]
SET Product_Price = NULL
WHERE ISNUMERIC(Product_Price) = 0;

UPDATE [Product Table]
SET Cost_Price = NULL
WHERE ISNUMERIC(Cost_Price) = 0;

----------------------------------------------------------------

-- 3. Replace negative or zero prices/costs with NULL
UPDATE [Product Table]
SET Product_Price = NULL
WHERE Product_Price <= 0;

UPDATE [Product Table]
SET Cost_Price = NULL
WHERE Cost_Price <= 0;

----------------------------------------------------------------
```

1. Fixing Structural Issues

Standardized column names and data types
Corrected invalid date formats
Split and merged inconsistent fields
Ensured proper mapping between products, categories, and territories

2. Removing Errors & Duplicates

Eliminated duplicate customer and order records
Removed repeated transaction lines
Corrected mismatched product IDs found in the early extraction stage

3. Handling Missing & Incomplete Data

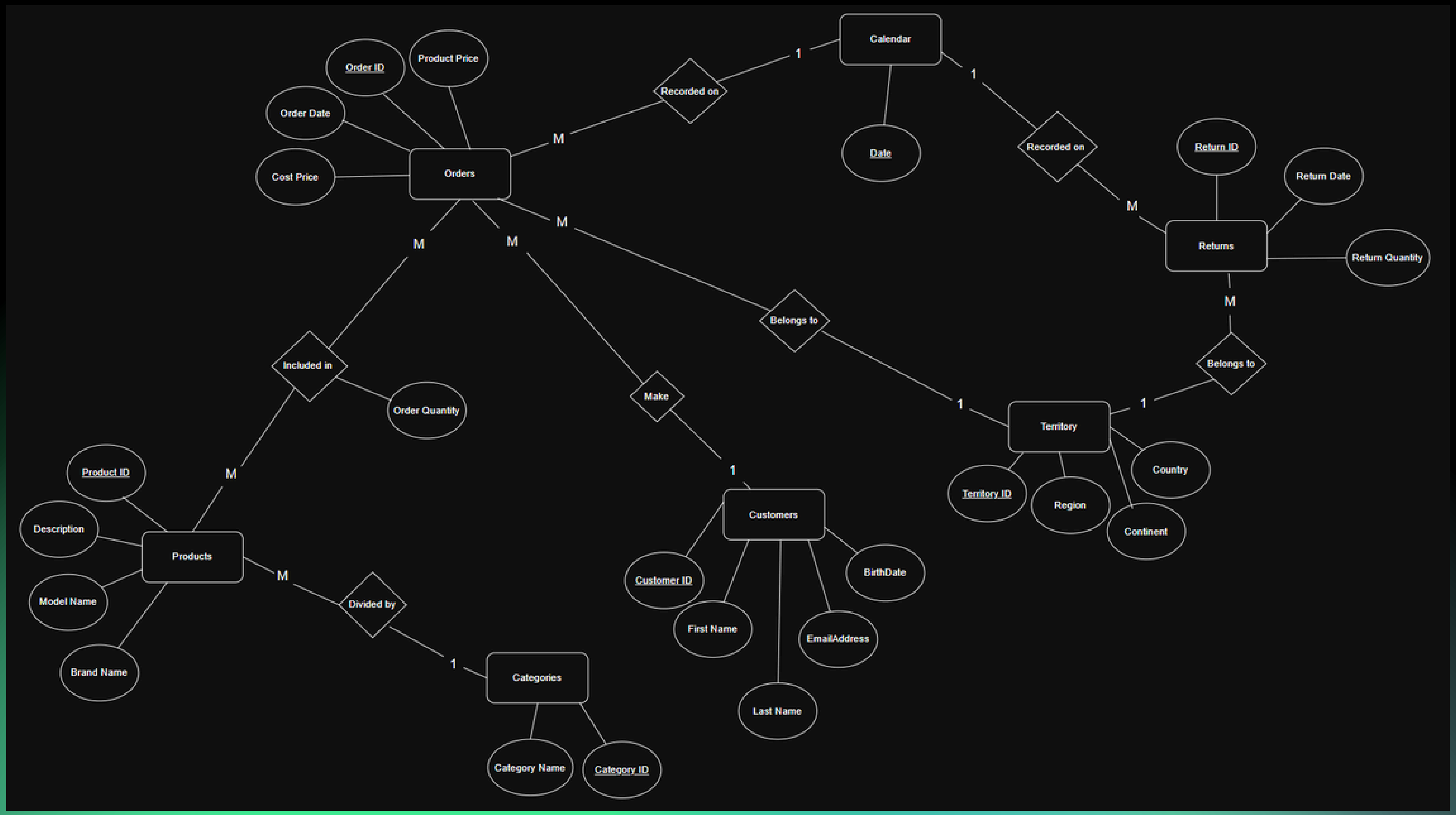
Filled or excluded records with missing values
Checked and repaired broken relationships
Validated returns data against actual orders

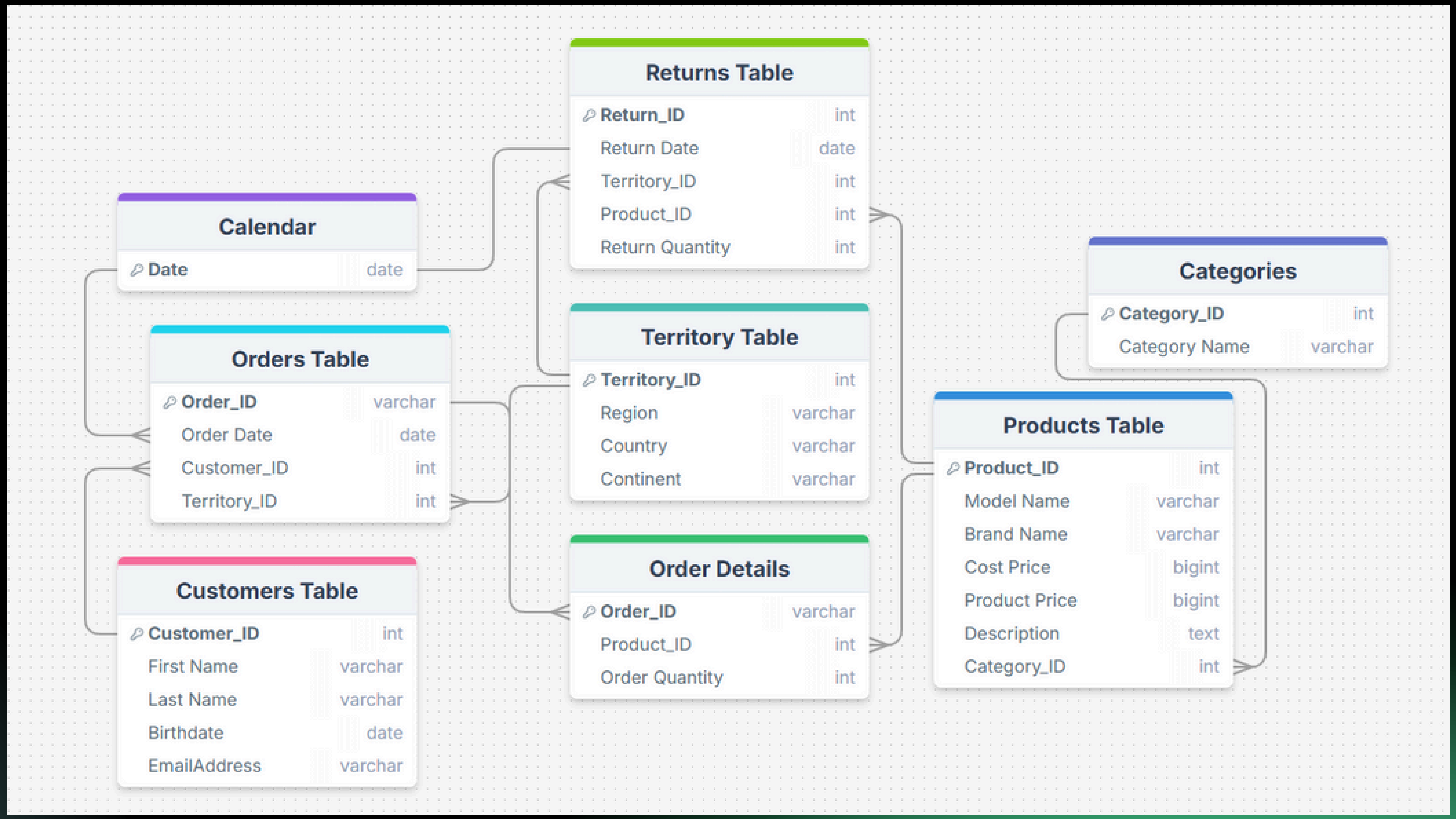
Initial Data Model & Mapping Schema

Before
DataWarehouse
Convert

Before building the data warehouse, we analyzed the original ERD to understand how the operational data was structured. The initial schema contained multiple tables with direct relationships between customers, products, orders, returns, and territories. These connections represented how the data existed in the source system, but they were not optimized for analytics.

The purpose of this step was to study the natural relationships, identify many-to-many issues, and detect missing keys or duplicated fields. This understanding helped us plan the transformation into a clean, structured, and analysis-ready star schema for the final dashboard.





Data Warehouse & Modeling

The Data Warehouse & Modeling phase was one of the most critical steps in transforming raw operational data into a structured, analytical system. The goal of this phase was to design a scalable, efficient, and accurate model that supports reliable business insights and advanced dashboard reporting.

1. Star Schema Design

We organized the data into a Star Schema with:

FactOrderDetails as the main fact table

Dimensions: Products, Categories, Customers, Territory, Calendar

This structure improved clarity and analytical performance.

2. Solving Relationship Issues

We removed the original Orders table to avoid duplication and created FactOrderDetails as the proper transactional table.

All relationships were set as one-to-many, ensuring a clean and stable model.

3. Time Intelligence Support

Using the Calendar dimension allowed accurate:

Monthly trends

Year comparisons

Time-based filtering

4. Model Validation

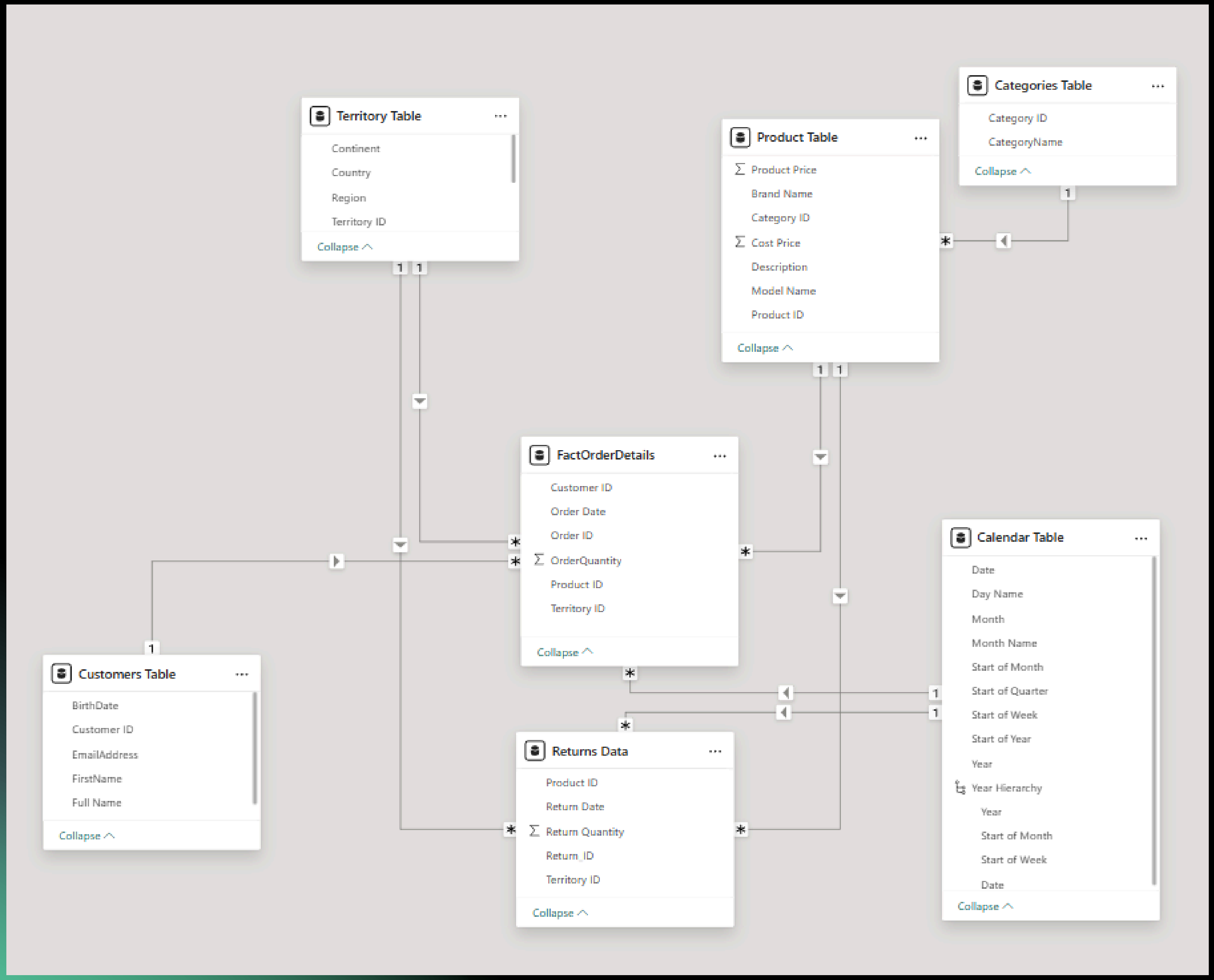
We validated:

Keys and relationships

Data consistency

Correct mapping between fact and dimensions

This ensured the warehouse was fully ready for insights and dashboard building.



DAX Measures & Calculated Logic

All DAX measures and calculated logic used in this project were fully created by the team based on the cleaned and prepared data model & the formulas were built manually inside Power BI using our own understanding of business requirements and analytical needs.

Measures Table

- Avg Profit Per Customer
- Avg Revenue Per Customer
- Customers Per Year
- Customers Up to Date
- Order Target
- Orders Gap
- Previous Month Orders
- Previous Month Profit
- Previous Month Returns
- Previous Month Revenue
- Profit Gap
- Profit Target
- Quantity Returned
- Quantity Sold
- Return Rate
- Revenue Gap
- Revenue Target
- Total Cost
- Total Customers
- Total Orders
- Total Products
- Total Profit
- Total Returns
- Total Revenue

1. Building KPI Measures

- Total Revenue
- Total Profit
- Total Cost
- Total Orders
- Total Products
- Total Customers
- Total Returns

2. Target & Gap Analysis

- Revenue Target
- Profit Target
- Order Target
- Revenue Gap
- Profit Gap
- Orders Gap

3. Time-Intelligence Measures

- Previous Month Revenue
- Previous Month Profit
- Previous Month Orders
- Previous Month Returns
- Customers Per Year
- Customers Up to Date

4. Quantity & Returns Logic

- Quantity Sold
- Quantity Returned
- Return Rate

5. Customer Value Insights

- Avg Profit Per Customer
- Avg Revenue Per Customer

Insights Extraction

```sql
-- 13.Who are the top customers according to the total revenue?
WITH CustomerSales AS (
    SELECT
        c.Customer_ID,
        CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
        SUM(f.OrderQuantity * p.Product_Price) AS TotalRevenue
    FROM FactOrderDetails f
    JOIN [Customers Table] c ON f.Customer_ID = c.Customer_ID
    JOIN [Product Table] p ON f.Product_ID = p.Product_ID
    GROUP BY c.Customer_ID, c.FirstName, c.LastName
)
SELECT *,
       RANK() OVER (ORDER BY TotalRevenue DESC) AS SalesRank
FROM CustomerSales
ORDER BY SalesRank;

-- 14.Who are the top 3 customers according to territory?
WITH TerritoryCustomers AS (
    SELECT
        t.Country,
        c.Customer_ID,
        CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
        SUM(f.OrderQuantity * p.Product_Price) AS TotalRevenue
    FROM FactOrderDetails f
    JOIN [Customers Table]c ON f.Customer_ID = c.Customer_ID
    JOIN [Product Table]p ON f.Product_ID = p.Product_ID
    JOIN [Territory Table] t ON f.Territory_ID = t.Territory_ID
    GROUP BY t.Country, c.Customer_ID, c.FirstName, c.LastName
)
SELECT *
FROM (
    SELECT *,
        ROW_NUMBER() OVER (PARTITION BY Country ORDER BY TotalRevenue DESC) AS Rank
    FROM TerritoryCustomers
) ranked
WHERE Rank <= 3
ORDER BY Country, Rank;

-- 15. Who are the inactive customers in last 3 months?
SELECT
    c.Customer_ID,
    CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
    MAX(f.Order_Date) AS LastPurchaseDate
FROM FactOrderDetails f
JOIN [Customers Table]c ON f.Customer_ID = c.Customer_ID
GROUP BY c.Customer_ID, c.FirstName, c.LastName
HAVING MAX(f.Order_Date) < DATEADD(MONTH, -3, GETDATE());
```

1. Sales Performance Insights

We analyzed overall sales trends across months, territories, and product categories to understand growth patterns and identify peak periods.

2. Customer Behavior Insights

We examined customer activity, purchase frequency, and distribution across regions to highlight strong customer segments and potential growth areas.

3. Product & Category Insights

We evaluated which products and categories generated the most sales and which ones performed weakly, helping us understand market demand and opportunities.

4. Returns & Loss Analysis

We studied returned orders to identify which products caused the most losses and how returns affected overall performance.

5. Actionable Findings

All insights contributed to practical recommendations for improving revenue, optimizing stock, reducing losses, and enhancing customer targeting.

Dashboard Creation

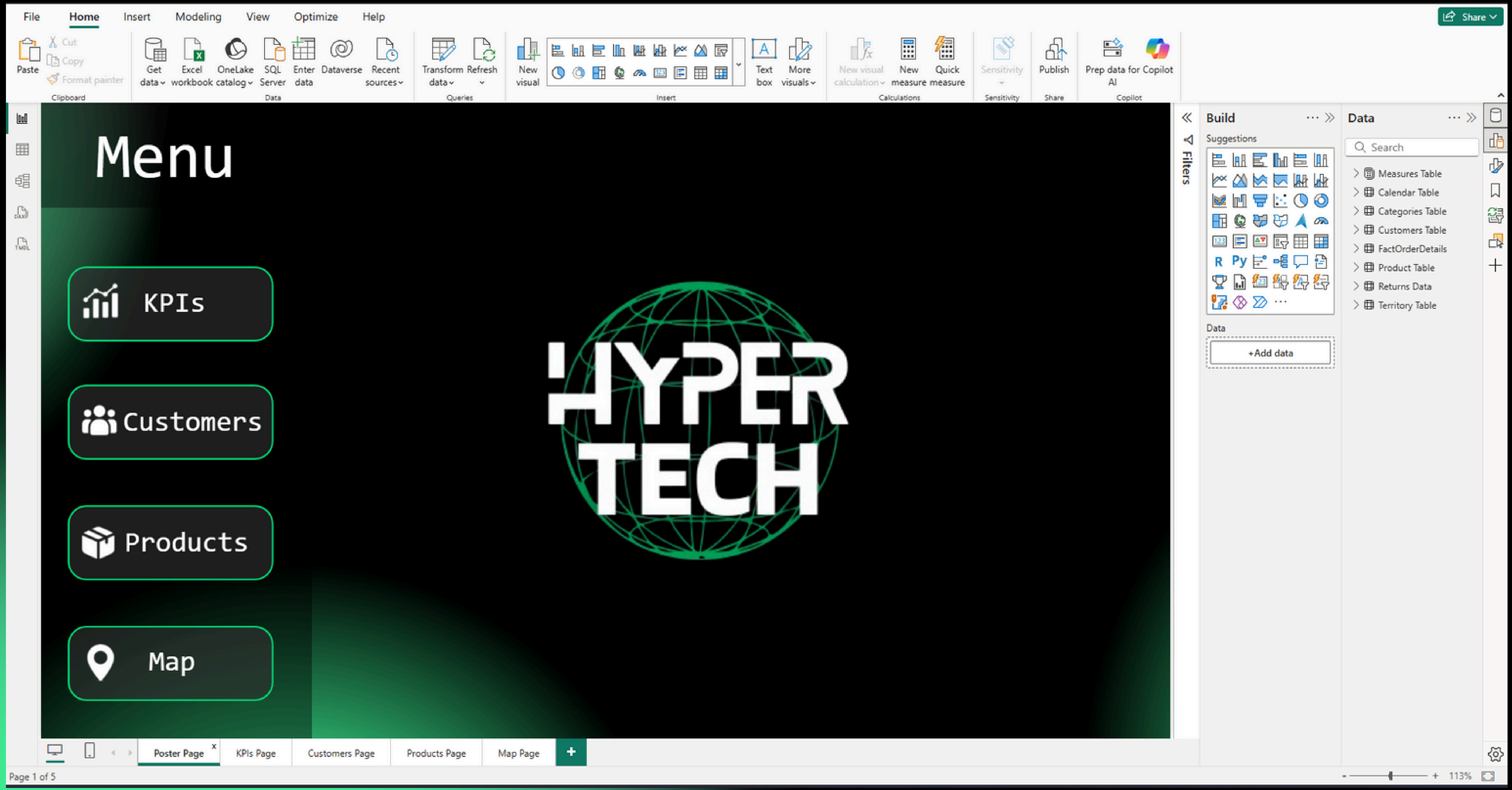
1. Connecting the Data Model
We connected the finalized Data Warehouse model to Power BI and Tableau, ensuring all relationships, keys, and measures worked smoothly across both tools.

2. Building Visual Analytics in Power BI
In Power BI, we created interactive visuals for:
Sales & profit trends
Customer distribution
Product & category performance
Return impact

3. Designing Tableau Dashboards
We developed a parallel dashboard short version in Tableau with clean layouts.

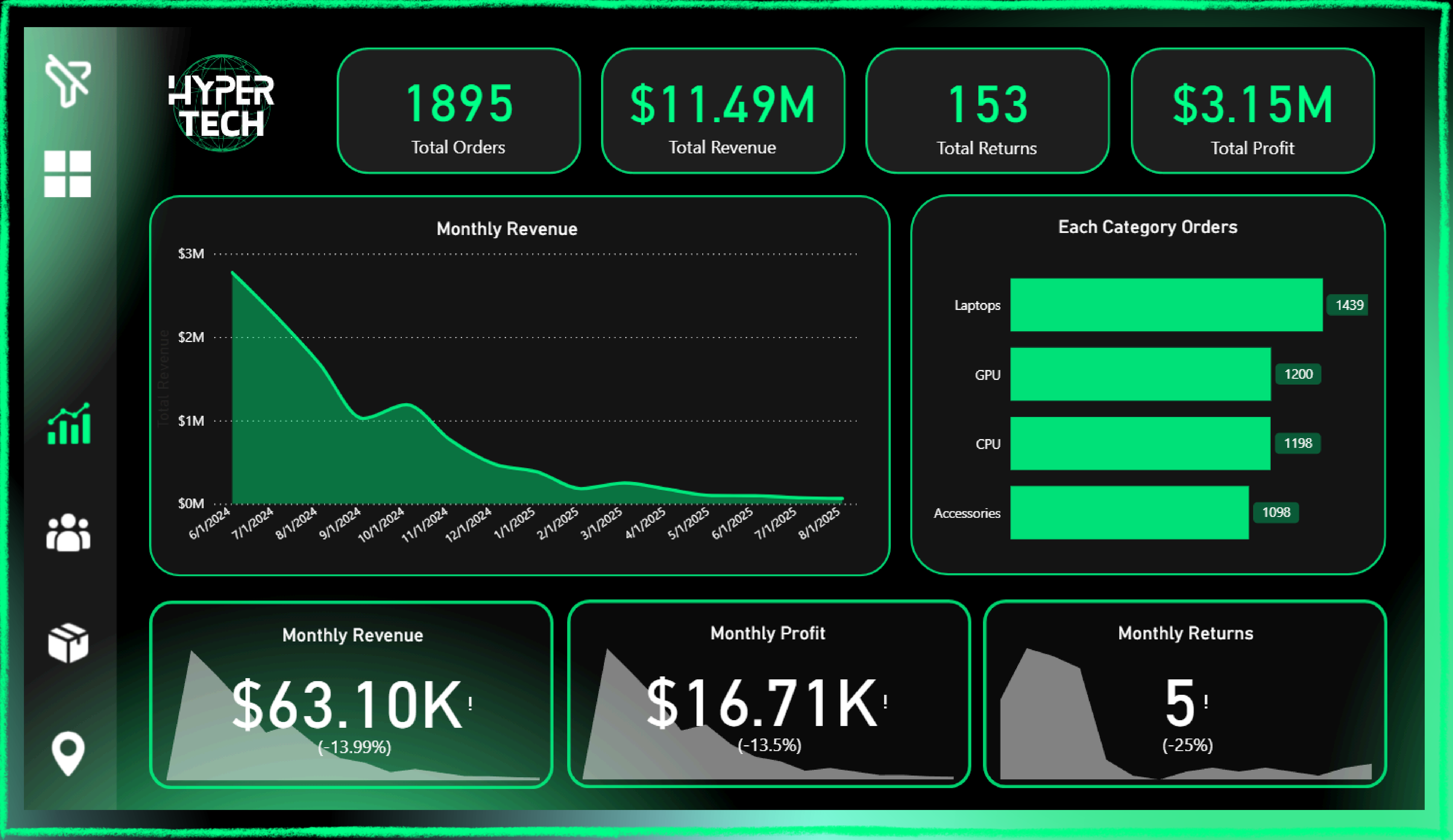
4. Applying Consistent Styling
We followed a unified design theme across all pages:
Consistent colors
Clear typography
Logical layout structure
High readability across all visuals

5. Ensuring Usability & Performance
We tested the dashboards to confirm:
Correct calculations
Fast loading time
Accurate filtering behavior
Smooth user interaction



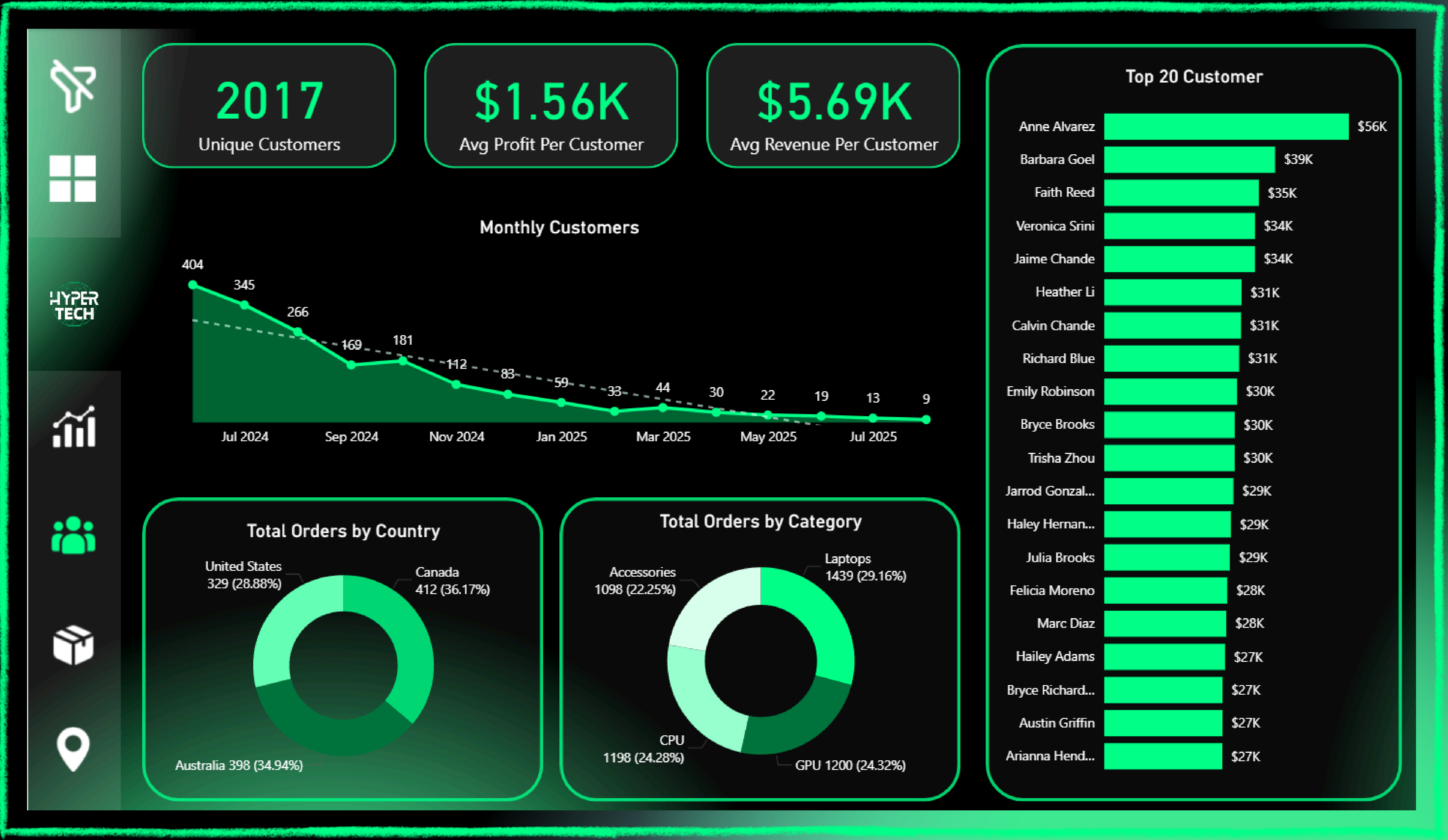
KPIs Page

The KPIs page summarizes the overall performance of Hyper Tech using the most important business indicators and gives an instant snapshot of the company's health and highlights trends in sales, returns, and profitability.



Customers Page

The Customers page focuses on understanding customer behavior and analyzing customer value and helps identify loyal customers, understand regional performance, and recognize high-value groups.

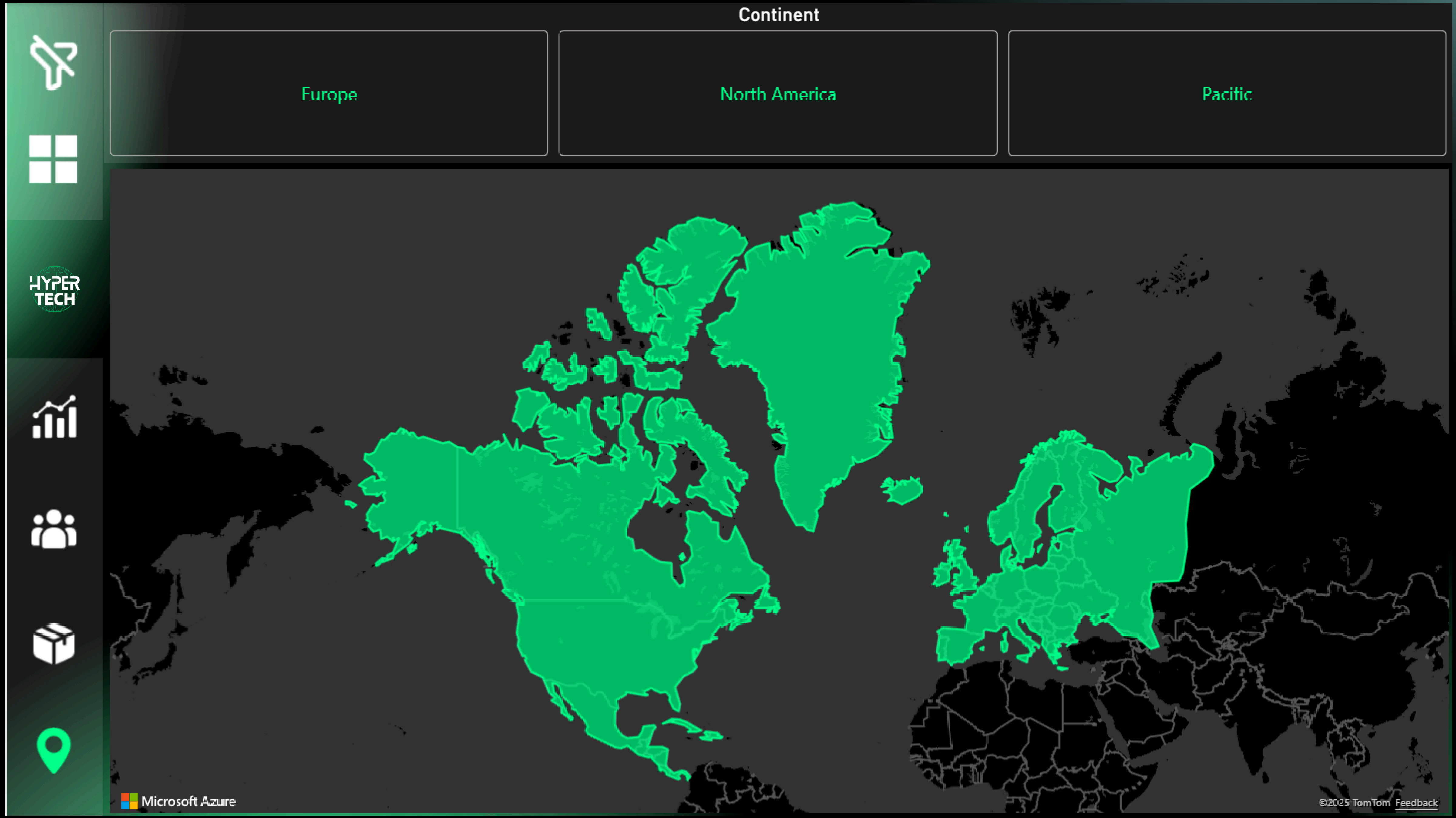


Products Page

The Products page analyzes the performance of product categories and brands, and also helps identify best-performing brands, monitor product categories, and locate items causing higher return activity.

Map Page

The Map page shows geographical performance across continents and countries and helps identify strong regions and markets that need improvement or growth.



Buttons & Filters

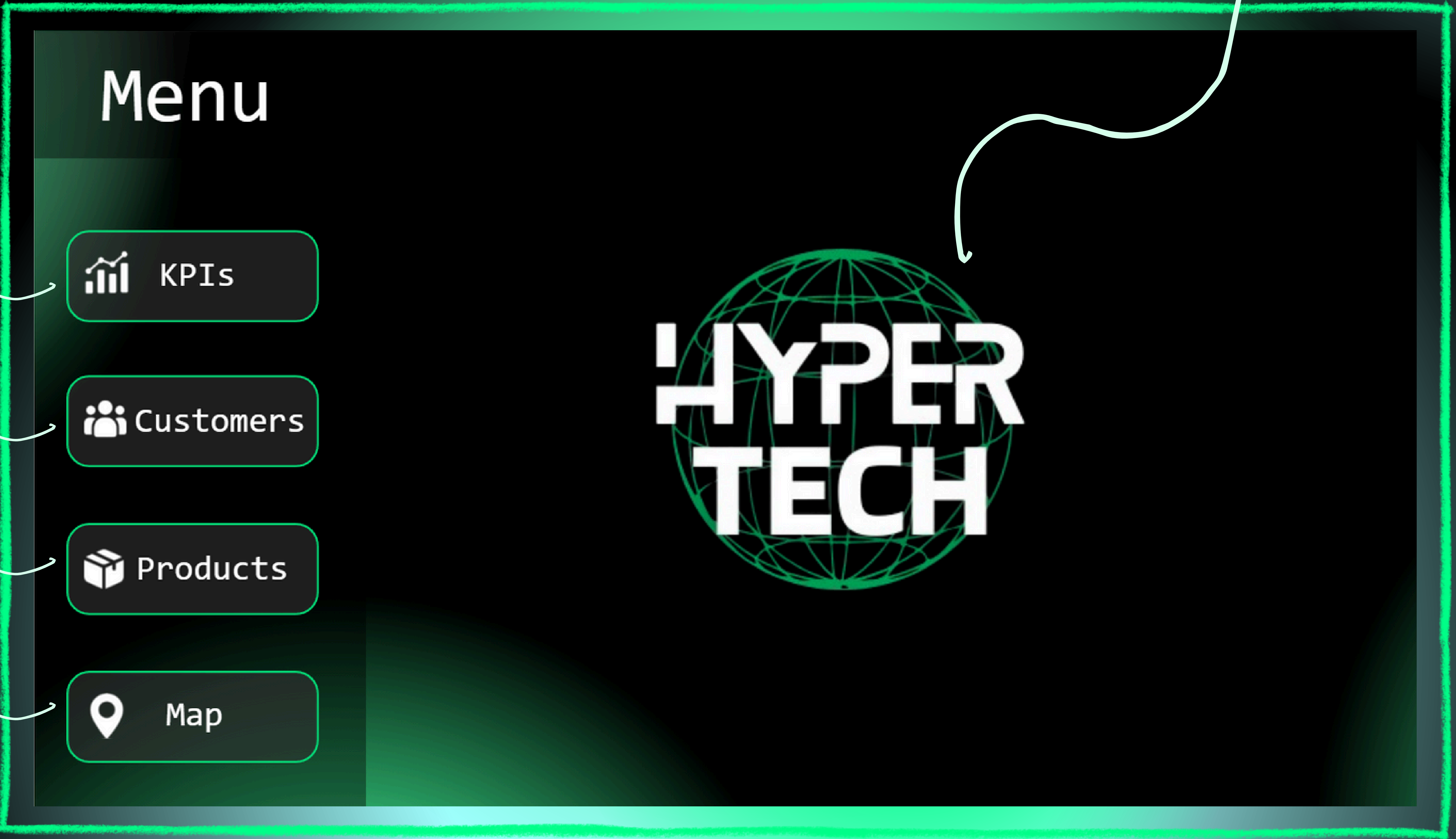
Hyper Tech Store Logo

KPIs Page Navigation

Customers Page Navigation

Prodcucts Page Navigation

Map Page Navigation



Buttons & Filters

Close Filters Menu

Year, Country and Category Filters

Clear Filters

Open Filters Menu

KPIs Page Navigation

Customers Page Navigation

Products Page Navigation

Map Page Navigation

Final Results & Business Impact

1. Complete Analytical Visibility
The dashboards now give Hyper Tech full visibility into sales, customers, products, territories, and returns.
Managers can evaluate performance quickly and accurately.

2. Stronger Data-Driven Decisions
The insights highlight top-performing products, weak categories, high-return items, and regional differences.
This helps the store understand where the strongest and weakest areas are.

3. Better Understanding of Customer Activity
Customer-focused KPIs reveal purchasing behavior, loyalty patterns, and regional demand.
This supports future decisions about targeting, promotions, and customer improvement strategies.

4. Clear View of Loss Sources
Return data now clearly shows where the financial loss is coming from, which products cause the most returns, and how these returns affect overall performance.
This prepares the store for developing future loss-reduction strategies.

5. Scalable BI Foundation
The structured model and dashboards can easily expand with new branches, new data sources, or future automation—giving Hyper Tech a long-term analytical system.

Strategies to Reduce Loss & Improve Business Performance

1. Improve Product Quality Control

Identify the items with the highest return rates
Communicate with suppliers to improve product quality
Apply quality checks before stocking new items
This reduces unnecessary returns and protects profit margins.

2. Optimize Inventory & Product Selection

Reduce stock of low-demand or high-return items
Increase investment in best-selling categories
Remove outdated or slow-moving products
Better inventory planning increases revenue and reduces losses.

3. Enhance Customer Support & After-Sales Service

Provide clear product information to avoid misunderstandings
Offer troubleshooting support before accepting returns
Train staff to help customers choose the correct products

4. Targeted Marketing Campaigns

Use customer segmentation (from the dashboard) to create personalized offers
Promote high-profit or best-selling brands
Use email and social media campaigns to re-engage inactive customers

5. Competitive Pricing & Promotions

Adjust prices based on category performance
Offer bundle discounts for related products
Create timed promotions during slow months
These strategies help increase sales and attract new customers.

6. Strengthen Supplier Partnerships

Negotiate better warranty policies
Request replacements for defective batches
Build relationships with reliable suppliers
This reduces the financial impact of returns.

7. Improve Website & Online Sales Experience

Highlight best-selling products
Improve product descriptions and specifications
Add customer reviews to build trust
A better online experience increases conversion rates and reduces return risk.

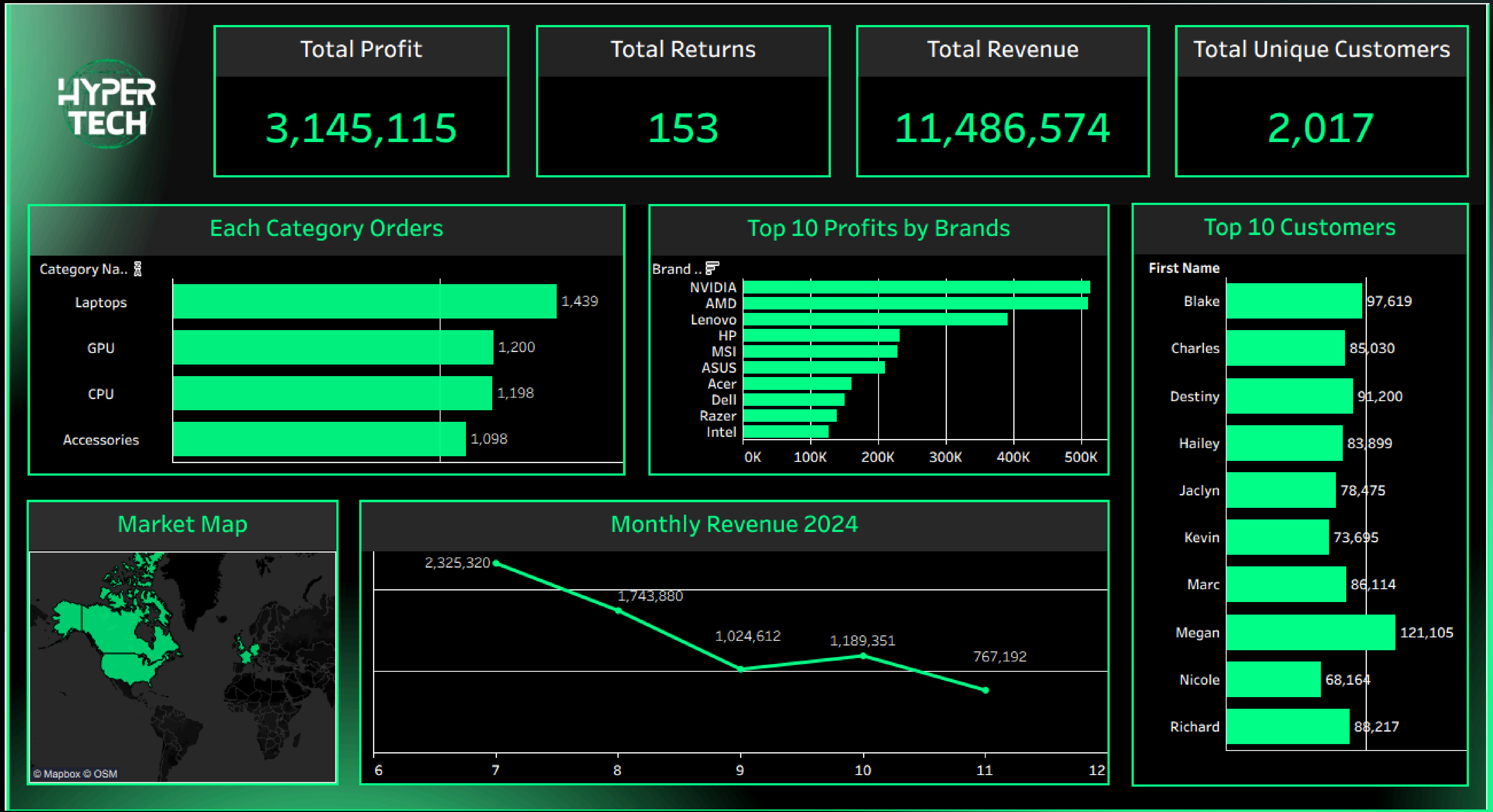
8. Use Data Insights to Guide Business Decisions

Leverage KPIs to track revenue, profit, and customer trends
Use the map page to invest more in high-performing regions
Monitor brand performance monthly to adjust stock levels

Tableau Dashboard Creation

This Tableau dashboard provides a simplified, high-level summary of Hyper Tech's performance. Unlike the detailed Power BI version, the Tableau dashboard focuses on presenting the essential insights in a clean and easy-to-read layout. It highlights the core KPIs, category performance, top brands, customer rankings, and monthly revenue trends in a compact format.

The purpose of this dashboard is to offer a fast, visual overview of the store's key metrics, allowing decision-makers to quickly understand overall performance without diving into deeper analytical layers. It serves as a lightweight companion to the full Power BI dashboard, making it ideal for presentations, quick reviews, and high-level monitoring.



Thank You

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